

Dong-A Univ. (ISPL)



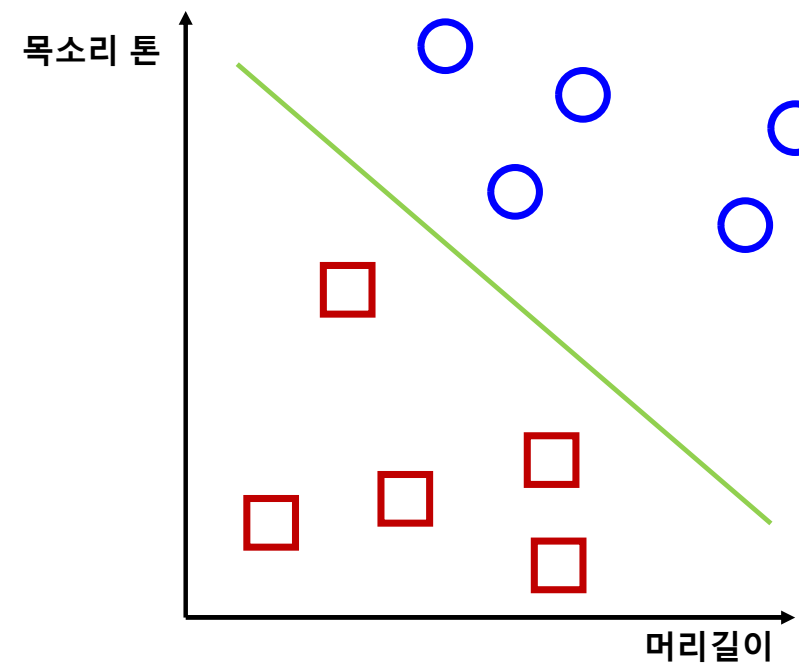
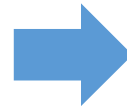
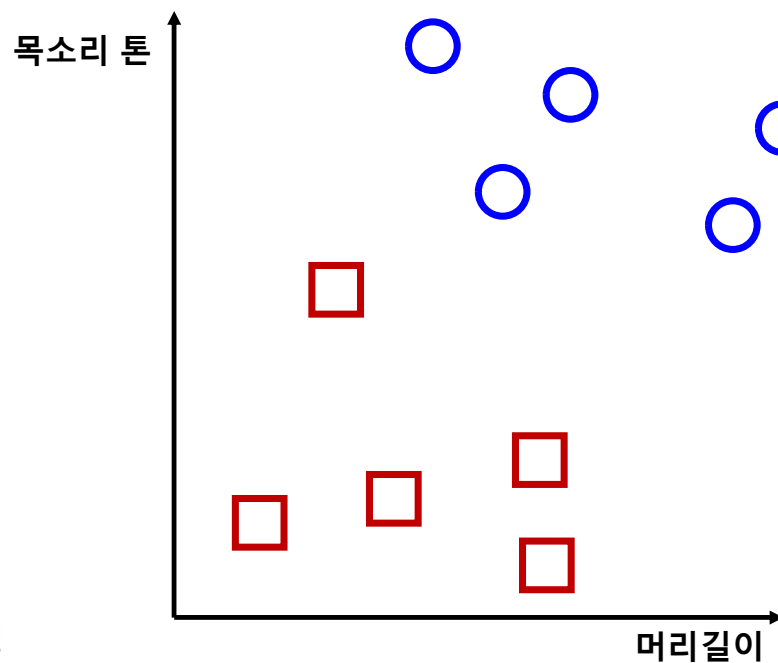
동아대학교
DONG-A UNIVERSITY

Support Vector Machine - 실습

컴퓨터공학부
2025년 1학기 머신러닝

Review – Support Vector Machine

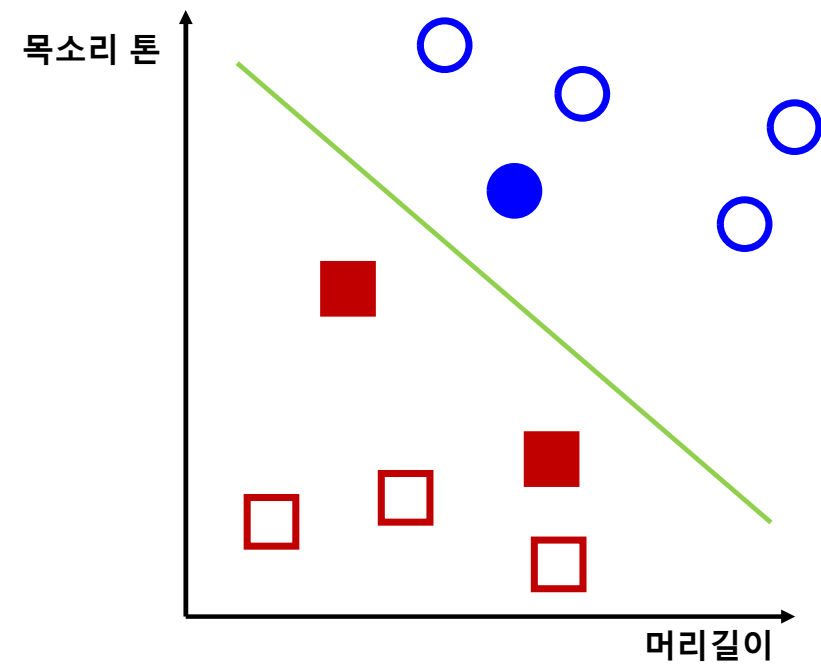
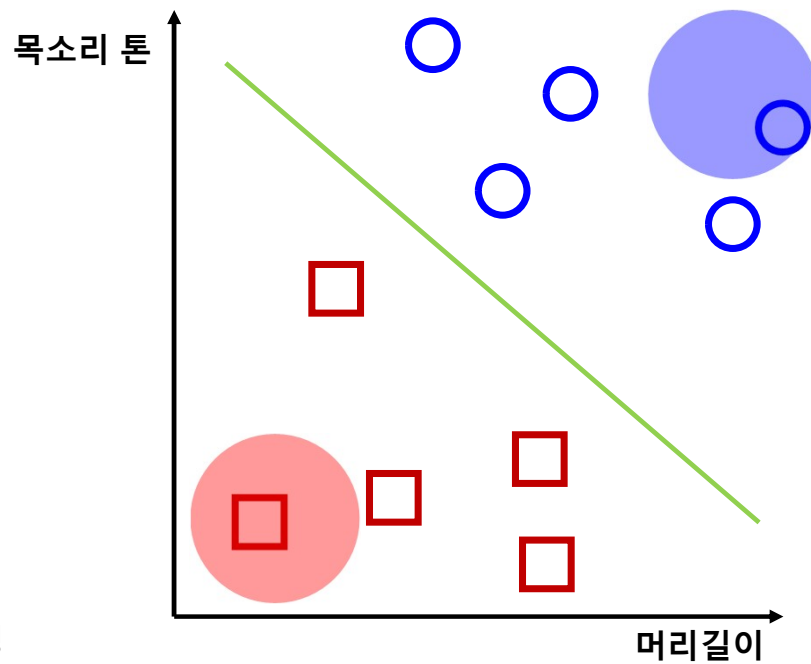
- 목적: Margin을 최대화하는 optimal separating hyperplane (decision boundary) 구하기



□ : 남성
○ : 여성

Review – Support Vector Machine

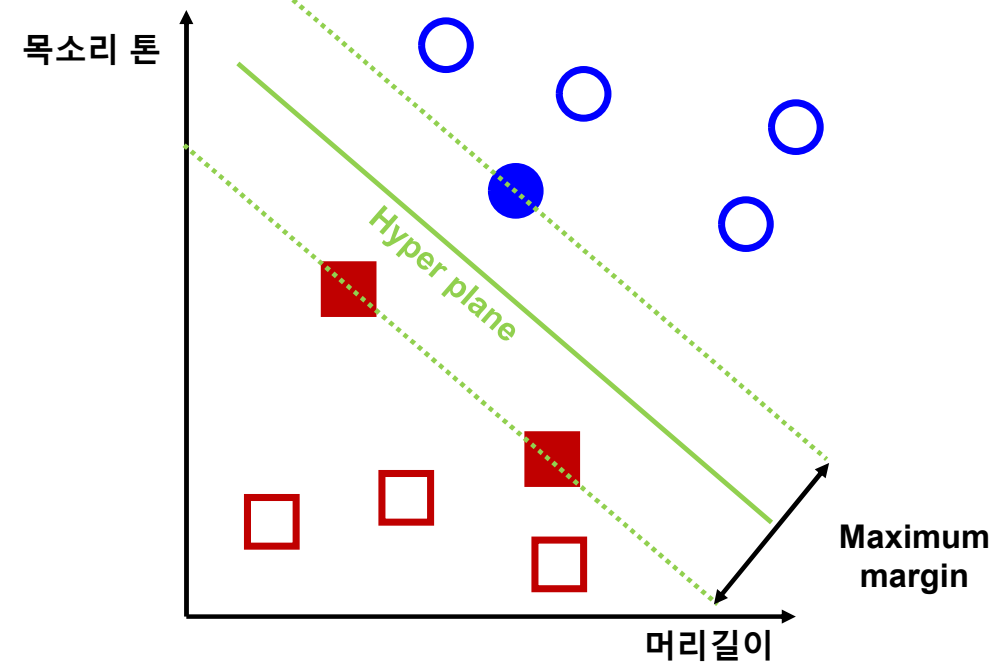
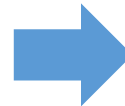
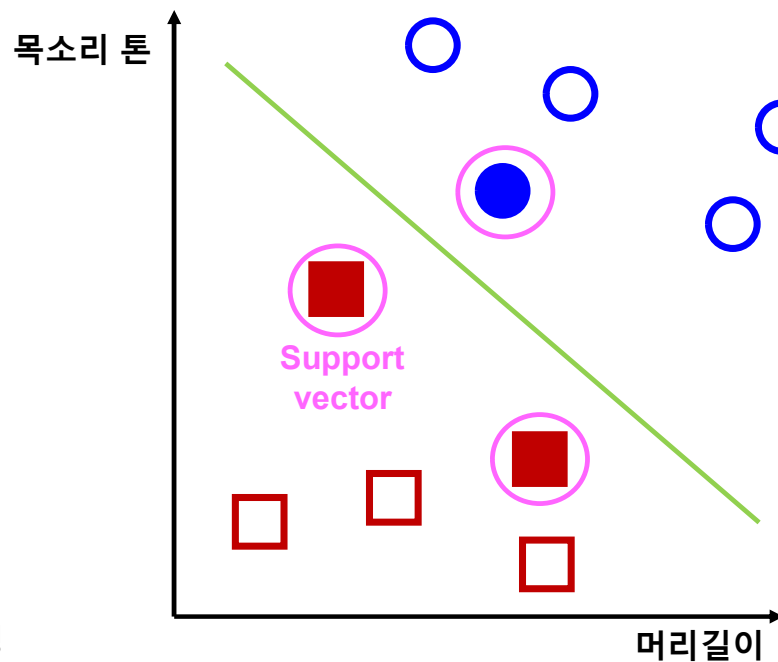
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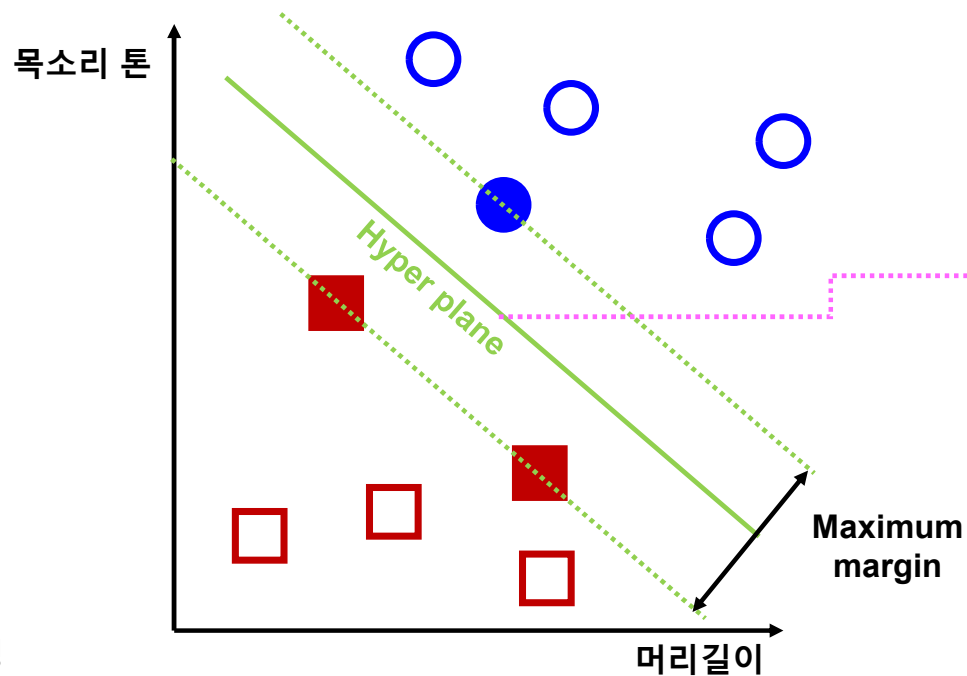
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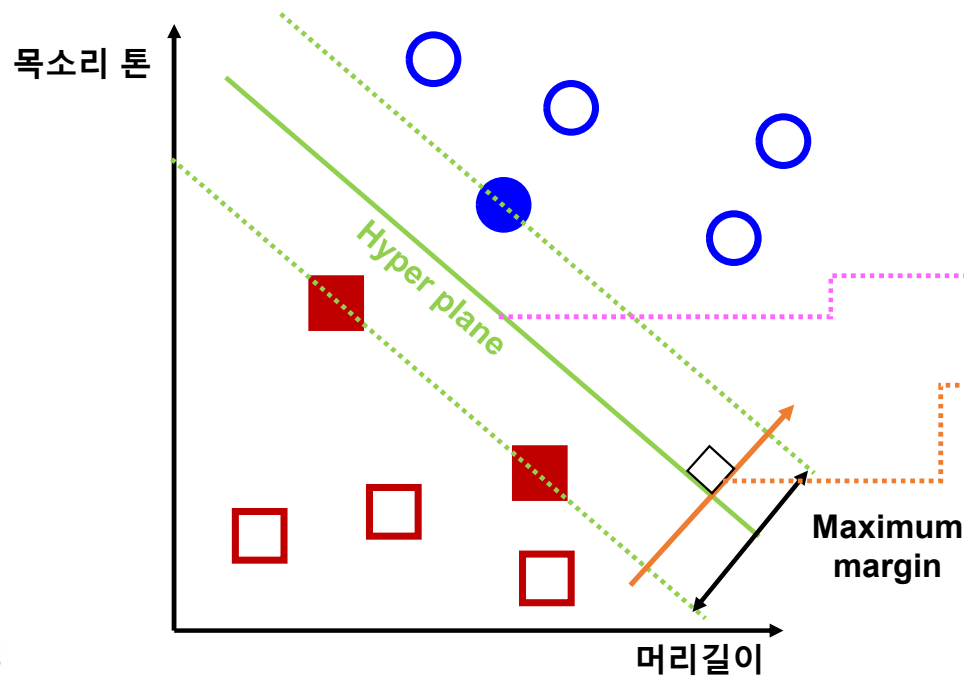


$$f(X) = W^T X + b$$

$$W^T X + b = 0 \text{ optimal separating hyperplane}$$

Review – Support Vector Machine

- 목적: Margin을 최대화하는 optimal separating hyperplane (decision boundary) 구하기



$$f(X) = W^T X + b$$

$$W^T X + b = 0 \text{ optimal separating hyperplane}$$

$$\vec{W} \begin{cases} \text{Weight (vector)} \\ \text{Normal vector} \end{cases}$$



Margin 방향의 unit vector

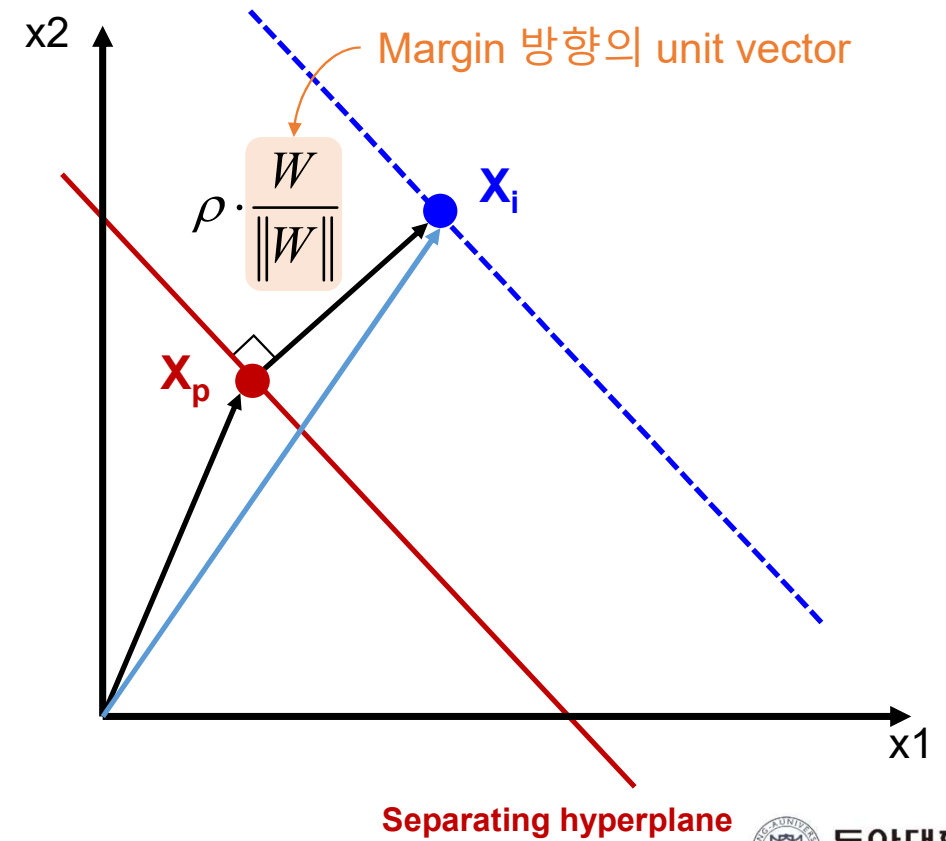
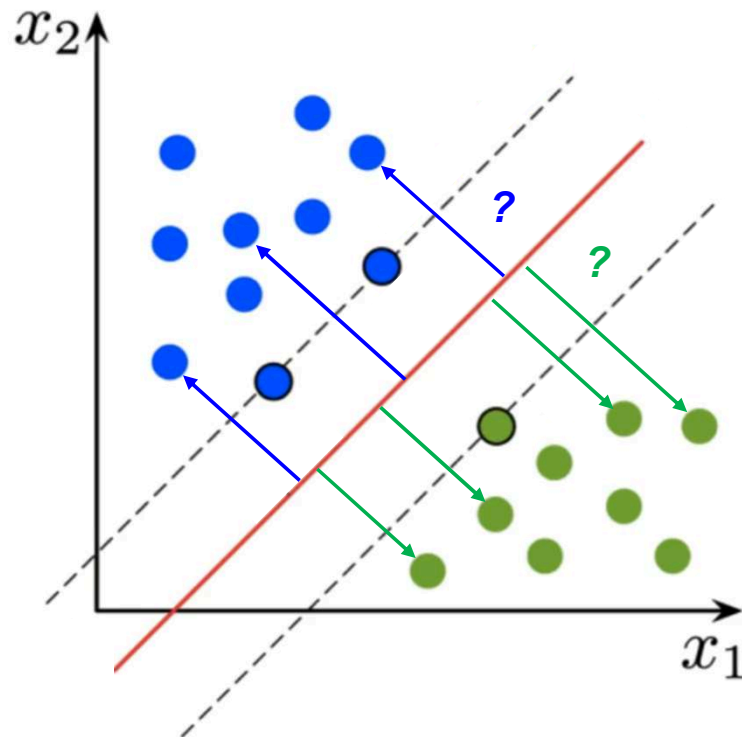
$$\frac{W}{\|W\|}$$

□ : 남성

○ : 여성

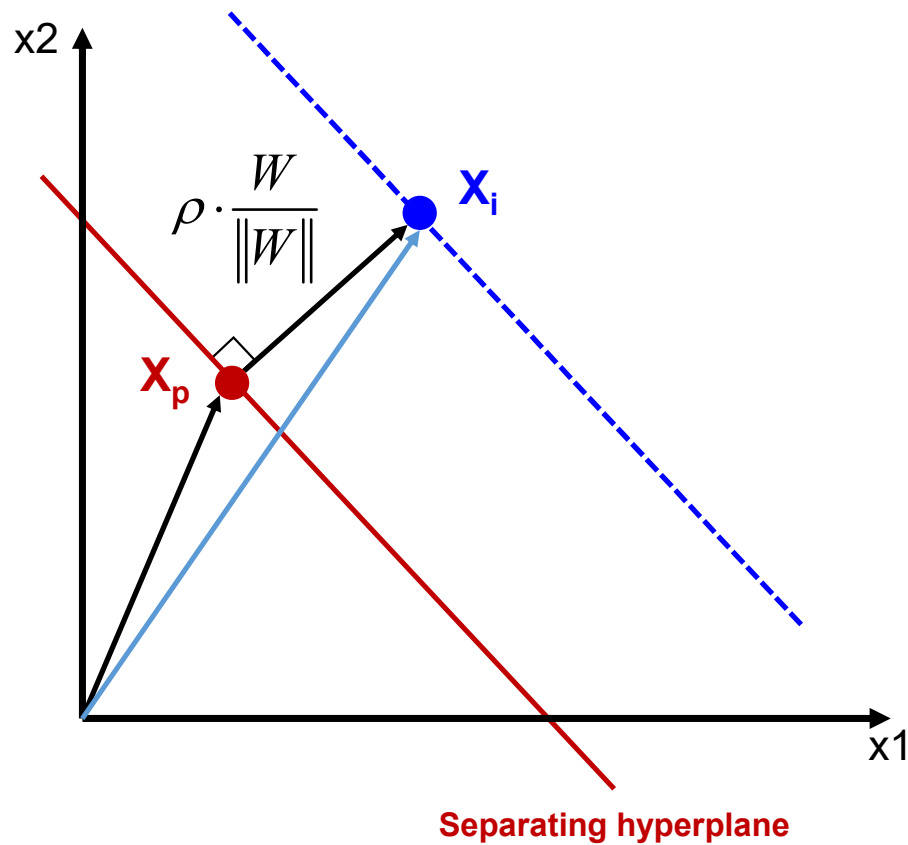
Review – Support Vector Machine

- 임의의 데이터 x_i 에 대해 separating hyperplane과의 거리: ρ

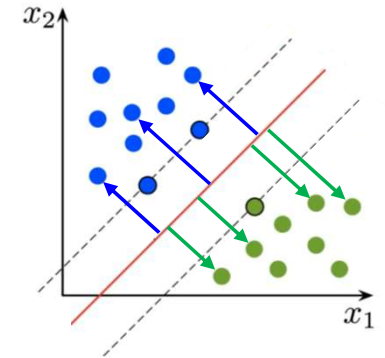


Review – Support Vector Machine

- 임의의 데이터 x_i 에 대해 separating hyperplane과의 거리: ρ



$$\begin{aligned}
 \textcircled{1} \quad & x_i = x_p + \rho \cdot \frac{W}{\|W\|} \\
 \textcircled{2} \quad & f(x_p) = W^T x_p + b = 0 \\
 \textcircled{3} \quad & f(x_i) = W^T x_i + b \\
 & = W^T \cdot \left(x_p + \rho \cdot \frac{W}{\|W\|} \right) + b \\
 & = \underbrace{(W^T x_p + b)}_0 + \rho \cdot \frac{\|W\|^2}{\|W\|}
 \end{aligned}$$



$$\textcircled{4} \quad \therefore \rho = \frac{f(x)}{\|W\|}$$

Review – Support Vector Machine

- Binary classification

1 : Optimal separating hyperplane

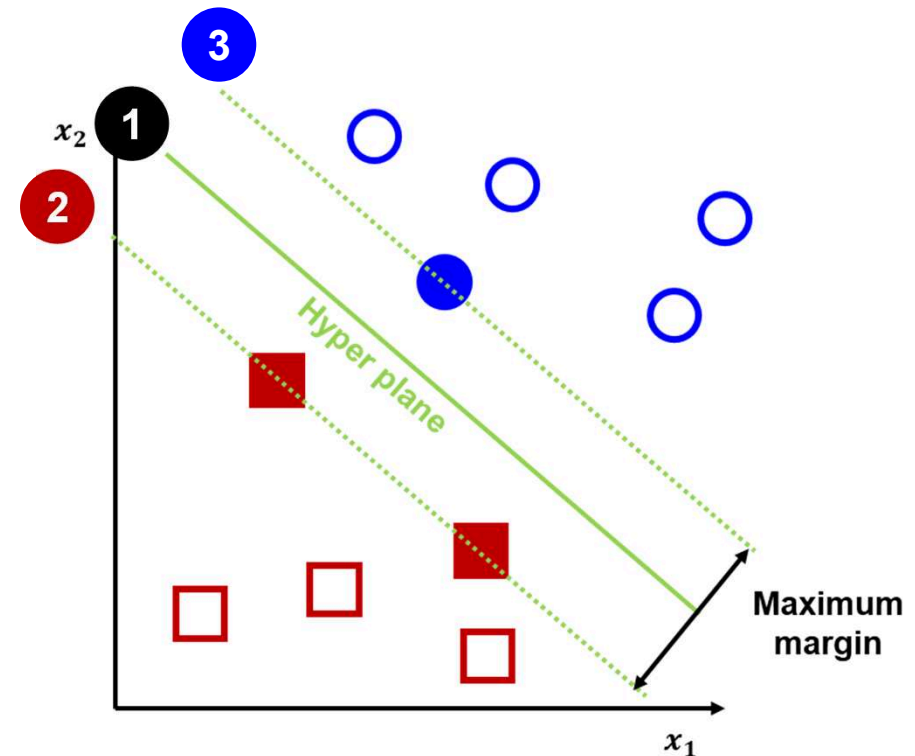
$$f(X) = W^T X + b = 0$$

2 : Support vector (negative)

$$f(X) = W^T X + b = -1$$

3 : Support vector (positive)

$$f(X) = W^T X + b = +1$$



Review – Support Vector Machine

▪ Binary classification

1 : Optimal separating hyperplane

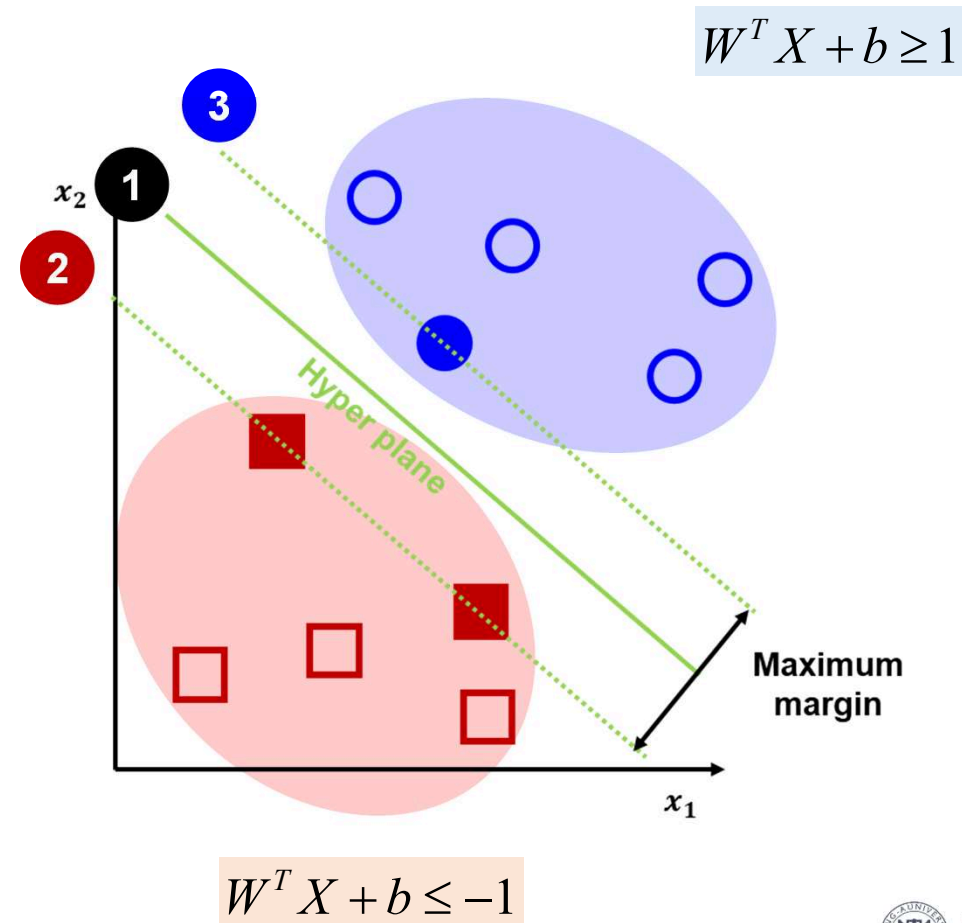
$$f(X) = W^T X + b = 0$$

2 : Support vector (negative)

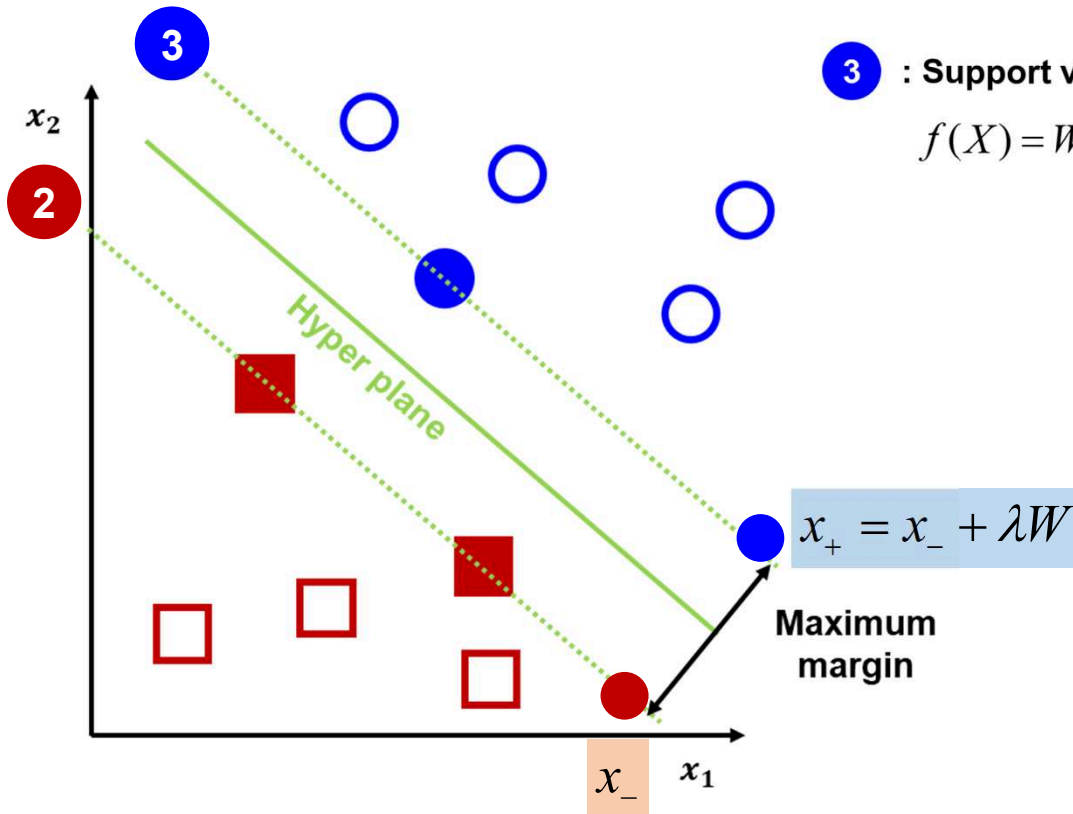
$$f(X) = W^T X + b = -1$$

3 : Support vector (positive)

$$f(X) = W^T X + b = +1$$



- **Margin 계산**

$$f(X) = W^T X + b = -1$$
$$f(X) = W^T X + b = +1$$


$$W^T x_+ + b = 1$$

$$W^T(x_- + \lambda W) + b = 1$$

$$W^T x_- + b + W^T \lambda W = 1$$

-1

$$\therefore \lambda = \frac{2}{W^T W}$$

- **Margin 계산**



3 : Support vector (positive)

$$W^T x_+ + b = 1$$

$$W^T(x_- + \lambda W) + b = 1$$

$$W^T x_- + b + W^T \lambda W = 1$$

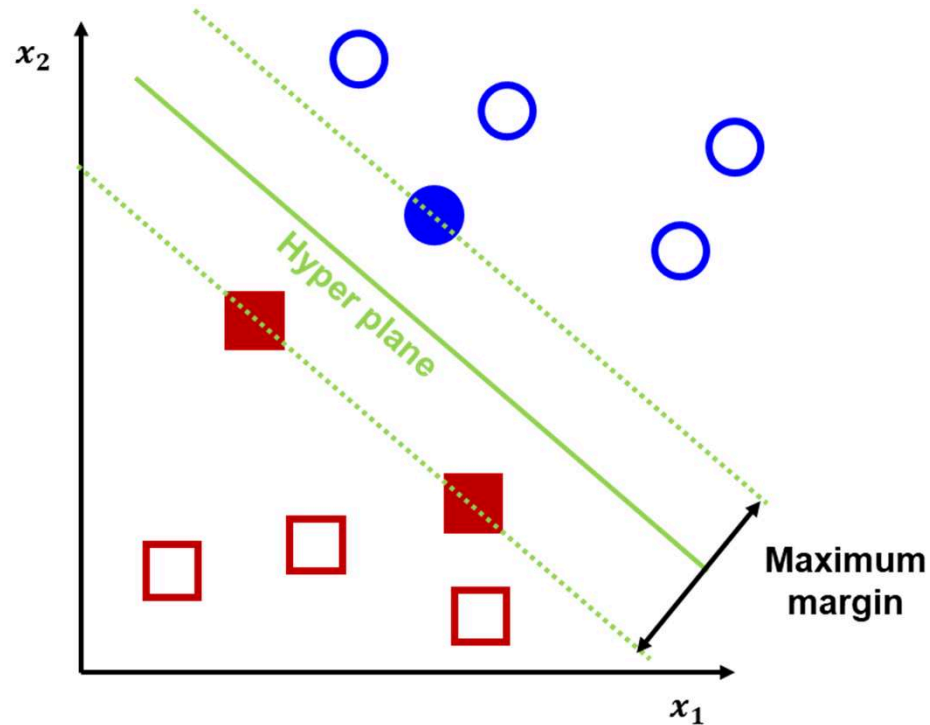
$$\therefore \lambda = \frac{2}{W^T W}$$

$$= \|x_+ - x_-\|_2 = \|\lambda W\|_2$$

$$= \frac{2}{W^T W} \cdot \sqrt{W^T W} = \frac{2}{\|W\|_2}$$

[실습] Support Vector Machine

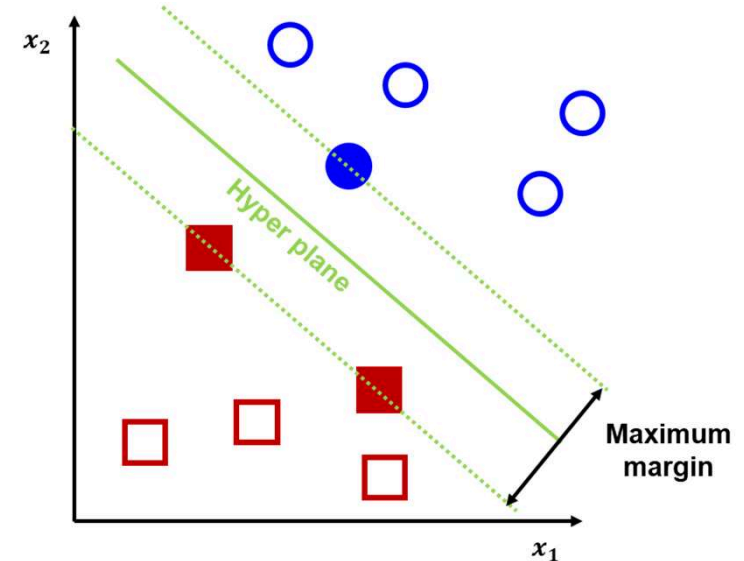
- 목적: Margin을 최대화하는 optimal separating hyperplane (decision boundary) 구하기



$$f(X) = W^T X + b$$

[실습] Support Vector Machine

- 목적: Margin을 최대화하는 optimal separating hyperplane (decision boundary) 구하기
- Solution
 - Gradient Decent Method (GD) → Optimal W, b
 - Quadratic Programming (2차 계획법)

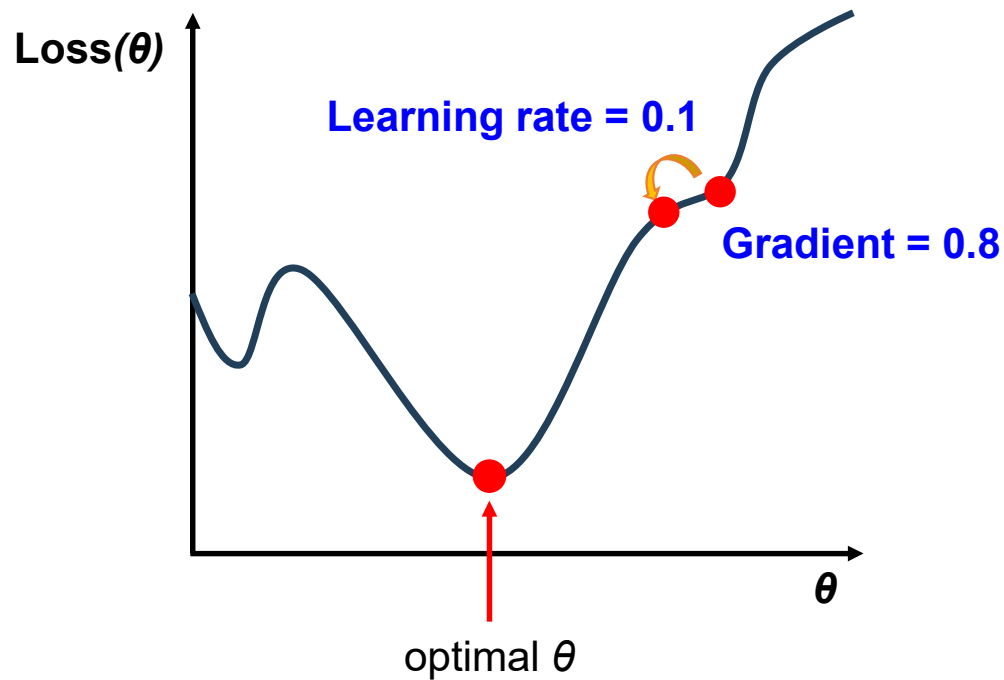


$$f(X) = W^T X + b$$

[실습] Support Vector Machine

■ Solution

- Gradient Decent Method (GD) → Optimal W, b
- Qudratic Programming (2차 계획법)



Gradient decent algorithm

- ① 현재 지점에서 미분을 이용해 gradient 계산
- ② Gradient에 learning rate를 곱하고 반대 방향으로 weight update

$$\begin{aligned}\theta_{t+1} &= \theta_t - \alpha \frac{\partial L}{\partial \theta_t} \\ &= \theta_t - 0.08\end{aligned}$$

[실습] Support Vector Machine

- Loss function (Cost function): Hinge loss

Prediction

$$W^T X_i + b \geq 1$$

→

Label

$$y_i = +1$$

$$W^T X_i + b \leq -1$$

→

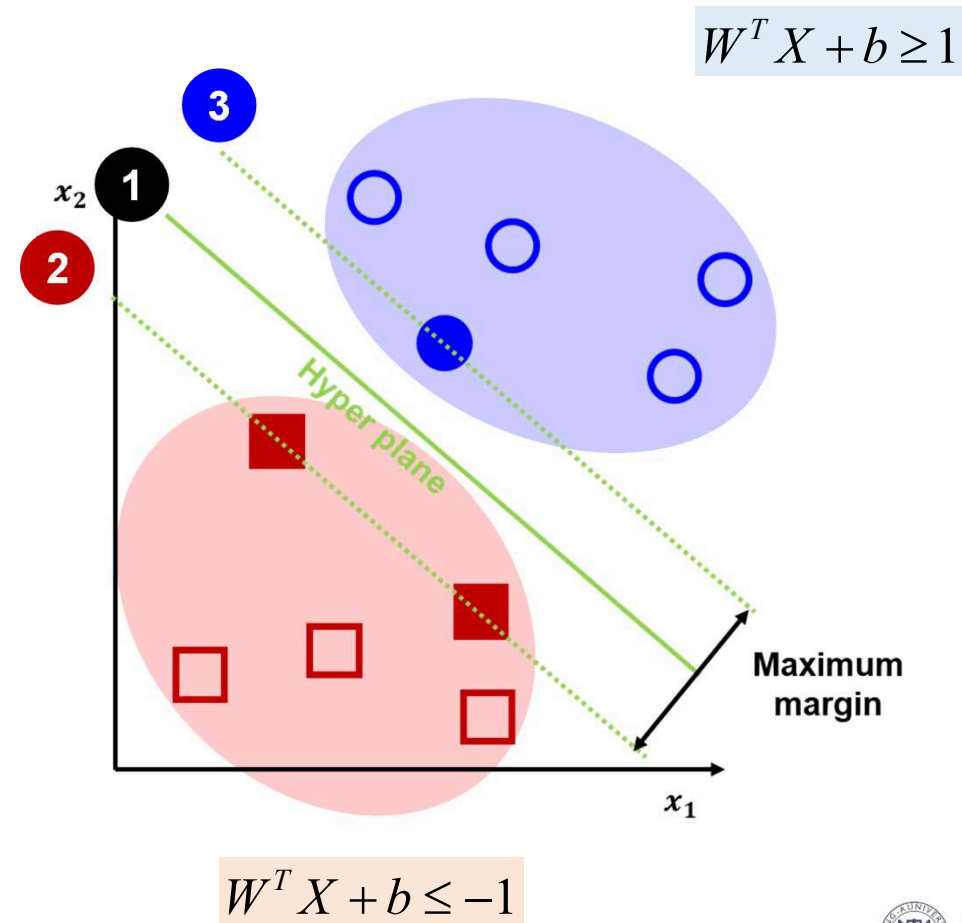
$$y_i = -1$$

$$y_i (W^T x_i + b) \geq 0$$

Label

Prediction

이 조건을 만족하는 경우 정상적으로 분류 성공



[실습] Support Vector Machine

- Loss function (Cost function): **Hinge loss**

Prediction

$$W^T X_i + b \geq 1$$

→

Label

$$y_i = +1$$

$$W^T X_i + b \leq -1$$

→

$$y_i = -1$$

$$y_i (W^T x_i + b) \geq 0$$

Label

Prediction

이 조건을 만족하는 경우 정상적으로 분류 성공

Hinge loss

$$Loss = \max(0, 1 - y_i (W^T x_i + b))$$

[실습] Support Vector Machine

- Loss function (Cost function): **Hinge loss**

Prediction

$$W^T X_i + b \geq 1$$

→

Label

$$y_i = +1$$

$$W^T X_i + b \leq -1$$

→

$$y_i = -1$$

$$y_i(W^T x_i + b) \geq 0$$

Label

Prediction

이 조건을 만족하는 경우 정상적으로 분류 성공

Hinge loss

$$Loss = \max(0, 1 - y_i(W^T x_i + b))$$

$$y_i(W^T x_i + b) = -1 \rightarrow Loss = +2$$

$$y_i(W^T x_i + b) = 0 \rightarrow Loss = +1$$

$$y_i(W^T x_i + b) = +0.5 \rightarrow Loss = +0.5$$

$$y_i(W^T x_i + b) = +1 \rightarrow Loss = 0$$

[실습] Support Vector Machine

- Loss function (Cost function): Hinge loss → Gradient

Hinge loss

$$Loss = \max(0, 1 - y_i(W^T x_i + b))$$

1 $y_i(W^T x_i + b) \geq 1 \implies Loss = 0$

2 *otherwise* $\implies Loss = 1 - y_i(W^T x_i + b)$

[실습] Support Vector Machine

- Loss function (Cost function): Hinge loss → Gradient

Hinge loss

$$Loss = \max(0, 1 - y_i(W^T x_i + b))$$

① $y_i(W^T x_i + b) \geq 1 \implies Loss = 0$

② *otherwise* $\implies Loss = 1 - y_i(W^T x_i + b)$

① $y_i(W^T x_i + b) \geq 1$

$$\frac{\delta L}{\delta W} = 0 \quad \frac{\delta L}{\delta b} = 0$$

Update 수행 X

② *otherwise*

$$\frac{\delta L}{\delta W} = -y_i x_i \quad \frac{\delta L}{\delta b} = y_i$$

[실습] Support Vector Machine

- Basecode 다운로드: LMS 강의 콘텐츠 13주차

Support Vector Machine (GD Method)

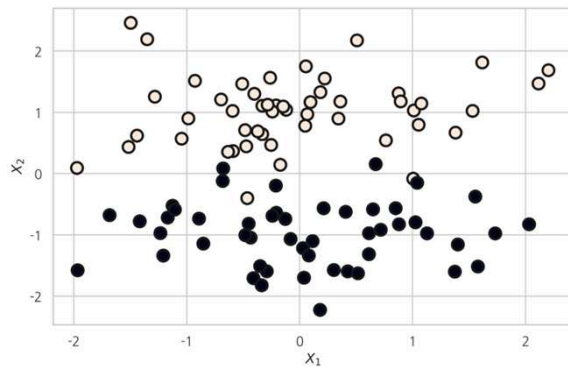
```
[ ] import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import make_blobs
```

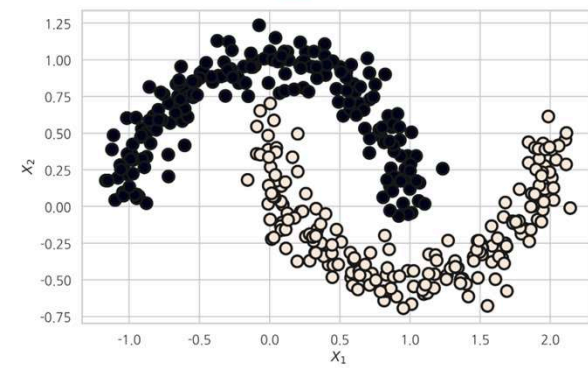
[실습] Support Vector Machine

- 데이터셋 생성: `sklearn.datasets`

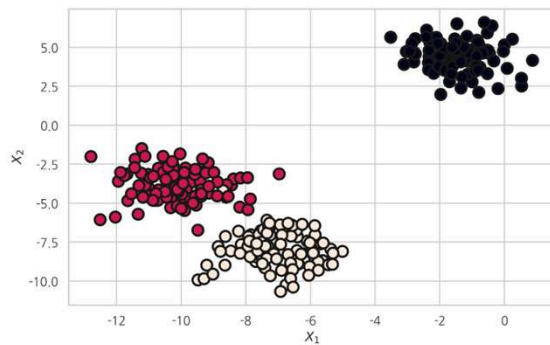
`make_classification`



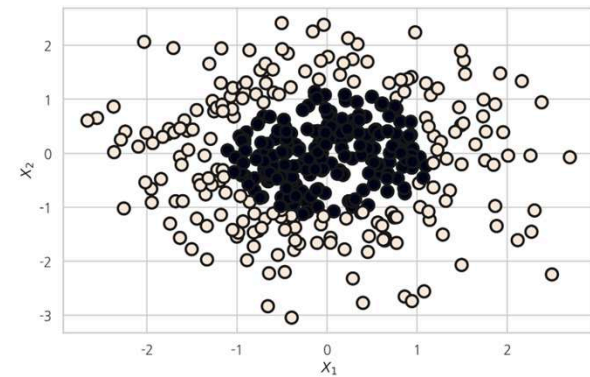
`make_moons`



`make_blobs`



`make_gaussian_quantiles`

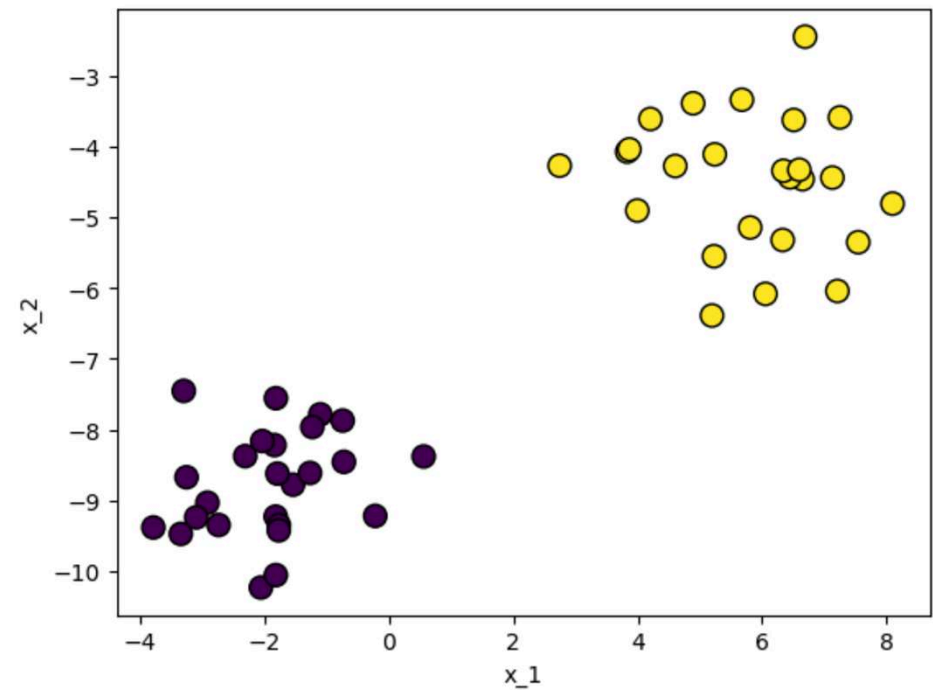


[실습] Support Vector Machine

■ 데이터셋 생성: sklearn.datasets

▼ Dataset

```
[ ] X, y = make_blobs(n_samples=50, n_features=2, centers=2, cluster_std=1.05, random_state=40)
plt.scatter(X[:, 0], X[:, 1], marker='o', c=y, s=100, edgecolor="k", linewidth=1)
plt.xlabel("x_1")
plt.ylabel("x_2")
plt.show()
```



[실습] Support Vector Machine

- SVM 모델 작성 및 gradient decent 코드 작성

Model

```
[ ] class SVM:
    def __init__(self, learning_rate=0.001, n_iters=1000):
        # initialization

    def fit(self, X, y):
        # Update parameters

    def predict(self, X):
        # Prediction
```

[실습] Support Vector Machine

- SVM 모델 training 및 도출된 W, b값 확인

Prediction

```
[ ] model = SVM()  
    margin_log = model.fit(X, y)  
  
    print(model.w, model.b)
```

```
[0.64070956 0.14828428] -0.125000000000000008
```

```
[ ] margin = 2 / np.sqrt(np.dot(model.w.T, model.w))  
    print(margin)
```

```
3.0411543613656318
```

$$\text{Margin} = \text{distance}(x_+, x_-)$$

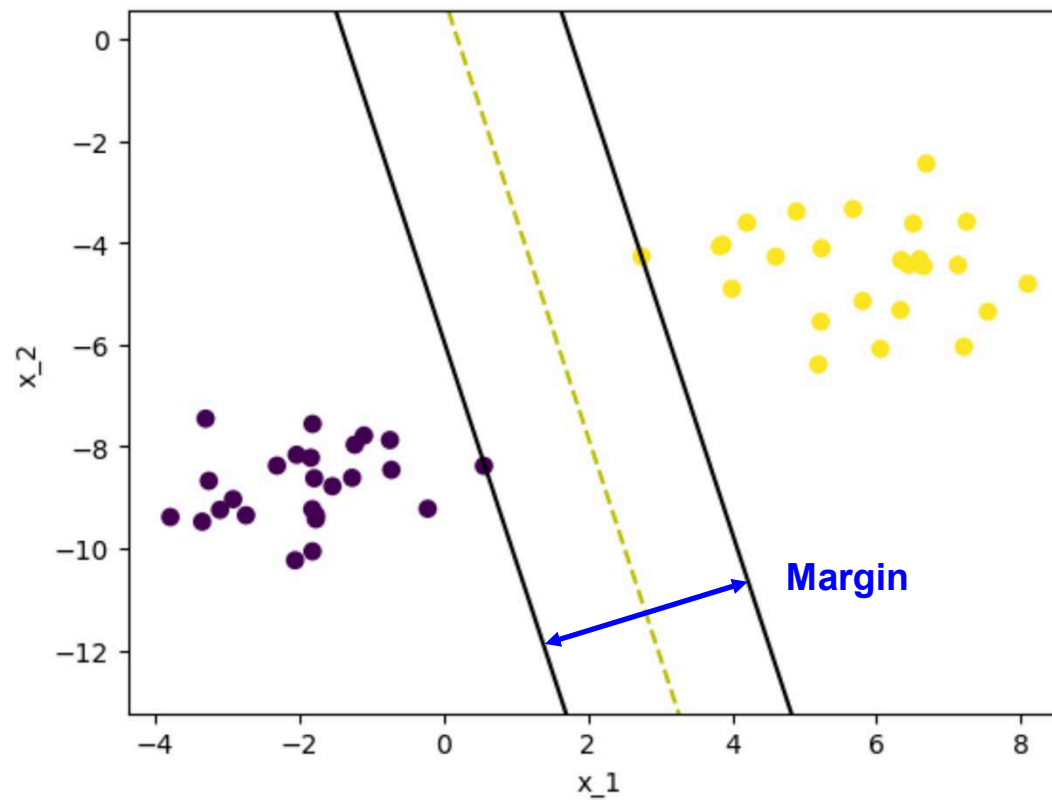
$$= \|x_+ - x_-\|_2 = \|\lambda W\|_2$$

$$= \frac{2}{W^T W} \cdot \sqrt{W^T W} = \frac{2}{\|W\|_2}$$

[실습] Support Vector Machine

Visualization

```
[ ] visualize_svm()
```



[실습] Support Vector Machine

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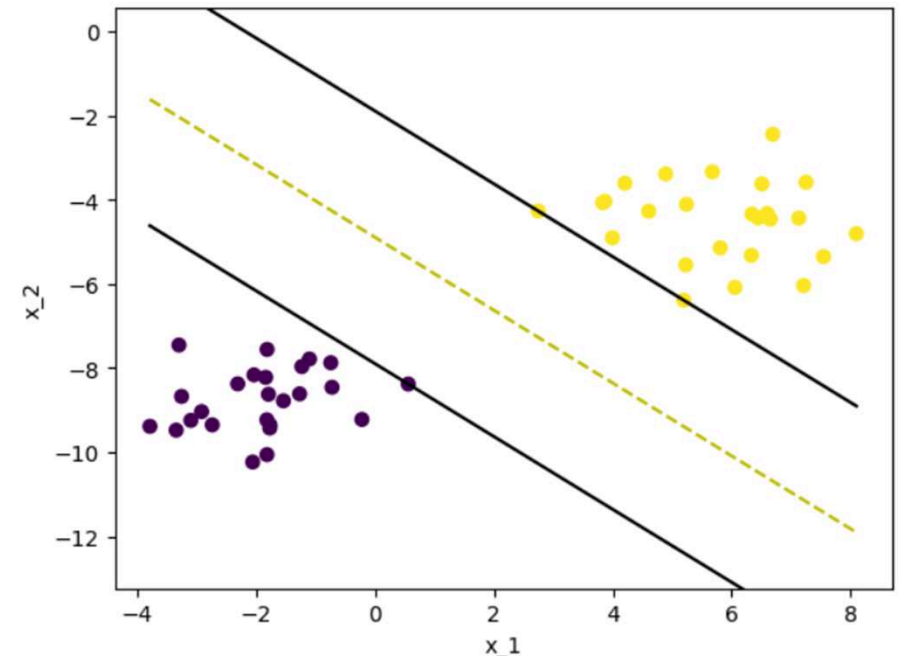
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✓ QP: Quadratic Programming 기반으로 SVM 최적화를 수행하는 scikit-learn 라이브러리

```
[14] from sklearn.svm import SVC
# 선형 커널을 사용하는 SVM 모델 정의 및 학습
clf = SVC(kernel='linear') # kernel: linear, rbf, poly, sigmoid 등
clf.fit(X, y)

# 시각화 실행
visualize_svm(clf.coef_[0], clf.intercept_)
```



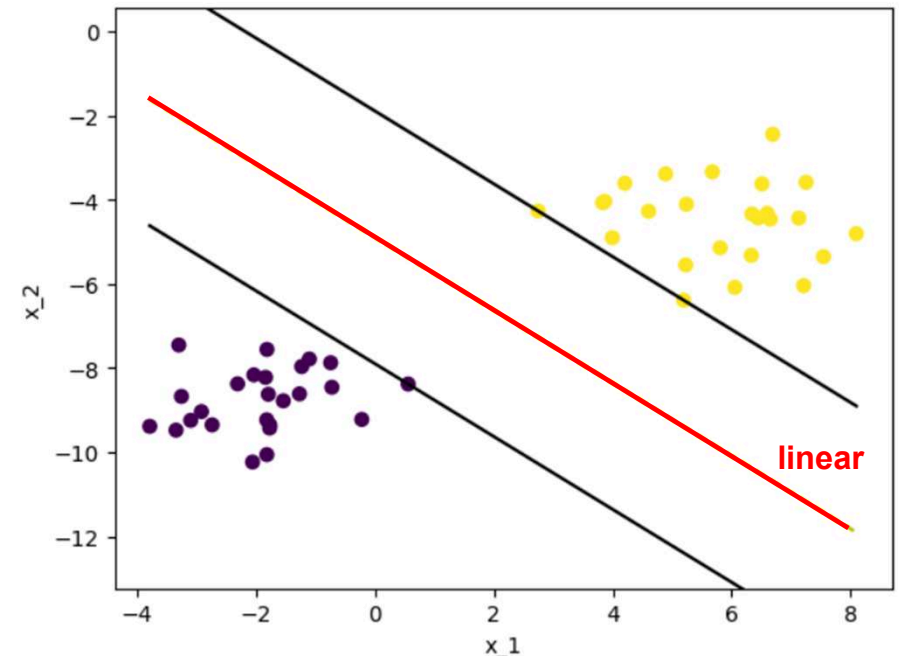
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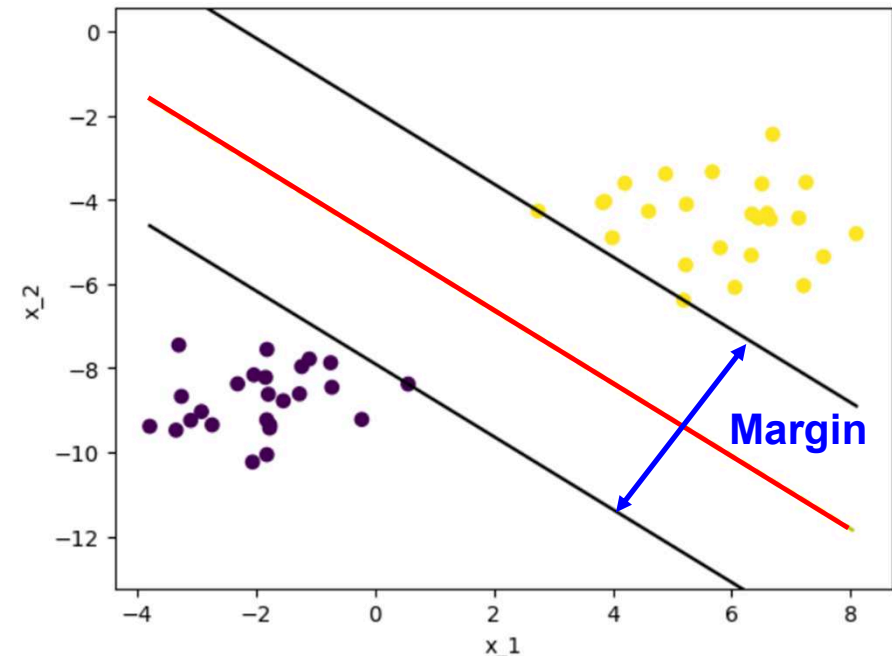
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fit 함수	Step1	커널 행렬 생성
	Step2	Quadratic Programming 문제 정의
	Step3	Convex Optimization
	Step4	$\lambda_i > 0$ 인 경우 support vector 선택
	Step5	w, b 계산





Questions & Answers

Dongsan Jun (dsjun@dau.ac.kr)

Image Signal Processing Laboratory (www.donga-ispl.kr)

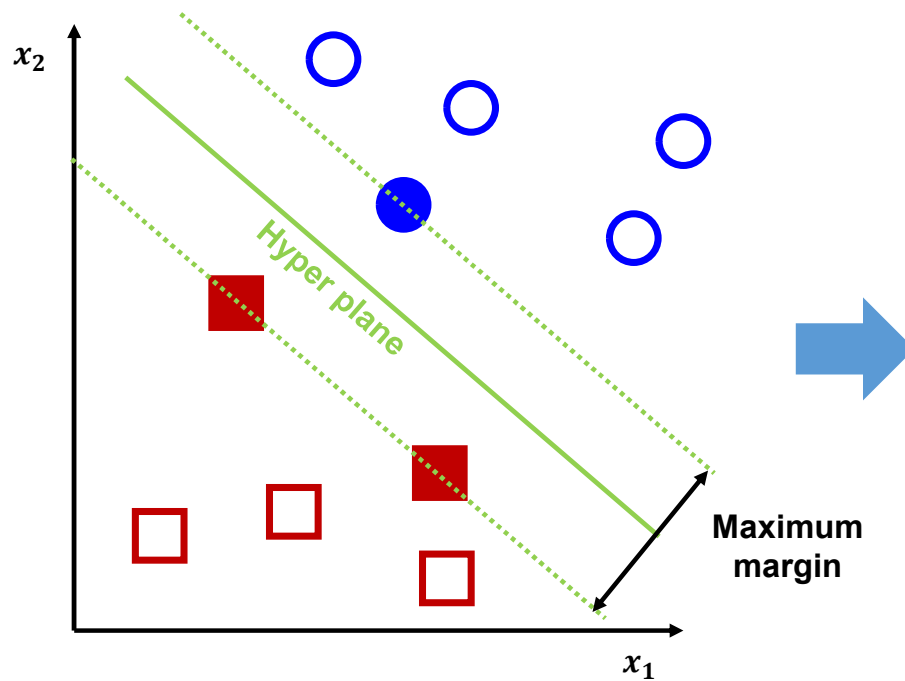
Dept. of Computer Engineering

Dong-A University, Busan, Rep. of Korea



Review – Support Vector Machine

- 목적: Margin을 최대화하는 optimal separating hyperplane (decision boundary) 구하기
- 예시: 스팸메일



□ : 스팸 메일

○ : 일반 메일

